# Lesson 2 Raspberry Pi Software installation and Test

## 2.1 Downloading the code program to control the robot

- For the power supply of Raspberry Pi, please refer to this official document:

  https://www.raspberrypi.org/documentation/hardware/raspberrypi/power/README.md

- Our Raspberry Pi robot driver board Motor HAT can directly supply power to the Raspberry Pi through the GPIO pins. However, because the software installation time in the Raspberry Pi is relatively long, it is not recommended to use battery power when installing the Raspberry Pi. The Raspberry Pi robot driver board Motor HAT or camera need not be installed when installing the software in the Raspberry Pi. This does not affect the software installation, but when you run the installed program, you must connect the driver board and the Raspberry Pi camera, otherwise Will cause the program to report an error.

- If you manually download the image file provided by us, you only need to load the SD card into the Raspberry Pi to boot, and the program of the robot product will run automatically. You can skip this section.

  Use the image file we provide If you want to update the code, you can delete the "**adeept_picar-b**" file and download the product code again.

  **sudo rm -rf adeept_picar-b**

  **sudo git clone https://github.com/adeept/adeept_picar-b.git**

- All the code of our product has been open sourced on GitHub, you need to download it to the Raspberry Pi and install the relevant dependencies before it can run normally. github address: https://github.com/adeept/adeept_picar-b

1. After the operations in the previous section, we have logged in to the Raspberry Pi, and enter the following commands in the console:

```
sudo git clone https://github.com/adeept/adeept_picar-b.git
```

2. After the input is complete, press enter to start downloading the robot program from GitHub. This process will continue for a period of time. Wait patiently for the download to complete.

Take MobaXterm as an example:

```
pi@raspberrypi:~ $
pi@raspberrypi:~ $ sudo git clone https://github.com/adeept/adeept_picar-b.git
Cloning into 'adeept_picar-b'...
remote: Enumerating objects: 188, done.
remote: Total 188 (delta 0), reused 0 (delta 0), pack-reused 188
Receiving objects: 100% (188/188), 3.85 MiB | 11.00 KiB/s, done.
Resolving deltas: 100% (59/59), done.
pi@raspberrypi:~ $
```

3. After the download is complete, a new folder" adeept_picar-b"will appear, in which the product code is stored. Check through the Linux command "ls".

```
ls
```

```
pi@raspberrypi:~ $
pi@raspberrypi:~ $ ls
adeept_picar-b          create_ap              Pictures                Templates
adeept_picarpro         Desktop                pocketsphinx-5prealpha  test
adeept_raspclaws        Documents              Public                  Videos
adeept_rasptankpro      Downloads              rpi-backup
adeept_trackingmodule   flask-video-streaming  sphinxbase-5prealpha
Bookshelf               Music                  startup.sh
pi@raspberrypi:~ $
```

## 2.2 Installing the dependency library of program

- If you use "Manually download the image file we provide" to write the image file to the SD card, the dependency library of the robot product has been installed, you can directly boot the Raspberry Pi to start using the robot.
- If you used "Download the official Raspbian image file" to write the image file to the SD card, you can refer to this section to install the dependent libraries.
- When installing dependent libraries, the installation of some dependent libraries may fail due to network problems or dependent library version issues. For dependent libraries that fail to install, the program will automatically install multiple times.
- After the installation is complete, the Raspberry Pi will automatically restart. At this time, the remote login program of the Raspberry Pi via SSH will be terminated. Please log in again.
- When installing dependent libraries, the installation may fail due to network or regional reasons. If you encounter any problems, please contact us: support@adeept.com

We have prepared a script to install all the dependent libraries that need to be used and set up operations such as turning on the camera and automatically running on startup.

Steps:

1. Enter the following code in the console and run the script setup.py to install the required dependent libraries:

**sudo python3 adeept_picar-b/setup.py**

```
pi@raspberrypi:~ $
pi@raspberrypi:~ $
pi@raspberrypi:~ $ sudo python3 adeept_picar-b/setup.py
Get:1 http://archive.raspberrypi.org/debian buster InRelease [32.8 kB]
Get:2 http://raspbian.raspberrypi.org/raspbian buster InRelease [15.0 kB]
0% [1 InRelease 6,809 B/32.8 kB 21%]
```

2. Press Enter to confirm. The following operations are automatically completed by the script program. Depending on the network environment, this process may last for tens of minutes or several hours, just wait patiently.

3. After the installation is complete, the console will show text:

```
The program in Raspberry Pi has been installed, disconnected and restarted.
You can now power off the Raspberry Pi to install the camera and driver board (R
obot HAT).
After turning on again, the Raspberry Pi will automatically run the program to s
et the servos port signal to turn the servos to the middle position, which is co
nvenient for mechanical assembly.
restarting...
```

After the installation is complete, the Raspberry Pi will automatically disconnect the SSH connection and restart. At this time, if you are using a Raspberry Pi connected by software such as Putty, there will be an error message such as Network error: Software caused connection abort, which is normal, just close the window.

## 2.3 Whether the dependent library is installed successfully

- When no error message appears when the "webServer.py" program is running, the dependent library is installed successfully.
- Only by successfully running the webServer.py program on the Raspberry Pi, can the Raspberry Pi be accessed with the IP on the browser.(After installing the dependent libraries, the Raspberry Pi will automatically run webServer.py).
- Prepare the components required for installation, and turn off the power of the Raspberry Pi during installation.

1. Install the camera cable, contact the metal surface of the cable with the metal surface of the Raspberry Pi (the same applies to installing the camera).

2. Install Motor HAT and camera, and connect the Raspberry Pi power supply.

3. After the Raspberry Pi is turned on (about 30-50s), you can access the Raspberry Pi with a browser.

- If the program is not run when the robot is booted, try to connect Raspberry Pi via SSH, manually run `webServer.py` with code and check the errors. Refer to the **Q&A** below or email us for help: support@adeept.com

- before manually running `webServer.py`, you need to end the program possibly auto run in the back end to release resources.

```
sudo killall python3
```

```
pi@raspberrypi:~ $ sudo killall python3
```

- Use the following command to run webServer.py

```
sudo python3 adeept_picar-b/server/webServer.py
```

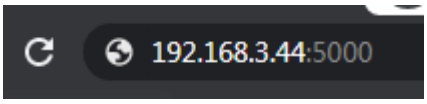
```
pi@raspberrypi:~ $ sudo python3 adeept_picar-b/server/webServer.py
.......................pause...........................
.......................pause...........................
Starting camera thread.
.......................pause...........................
 * Serving Flask app "app" (lazy loading)
 * Environment: production
   WARNING: Do not use the development server in a production environment.
   Use a production WSGI server instead.
192.168.3.98
 * Debug mode: off
waiting for connection...
 * Running on http://0.0.0.0:5000/ (Press CTRL+C to quit)
```

- Open the web control interface.

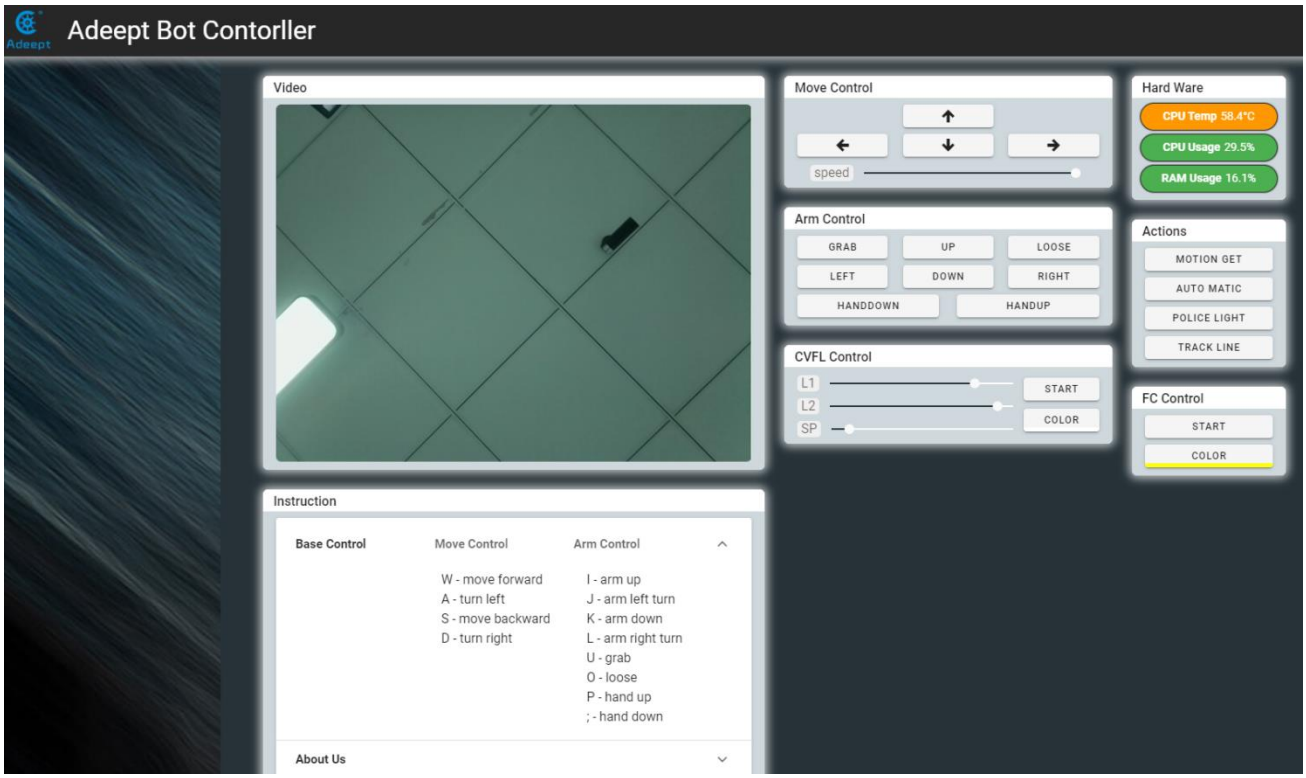
1. Make sure your device is in the same local area network as the Raspberry Pi.

2. Obtain the IP address of the Raspberry Pi (refer to the software installation section).

3. Open the browser on the device (chrome browser is recommended to avoid possible browser compatibility issues), enter the IP address of your Raspberry Pi in the address bar, and visit port 5000, for example: 192.168.3.44:5000




4. The web controller will then be loaded into the browser.




Depending on the product, the modules on the web controller are also different.

## 2.4 Creating a WiFi Hotspot on Raspberry Pi(Less used)

The dependent library is successfully installed, and after "webServer.py" runs automatically, our product will automatically turn on the hotspot when the wifi is not connected.

The method of turning on the WIFI hotspot in our robot product uses a project from GitHub create_ap. Our installation script will automatically install this program and related

dependent libraries. If you have not run our installation script, you can use the following command to install create_ap:

```
sudo git clone https://github.com/oblique/create_ap.git
```

```
cd create_ap
```

```
sudo make install
```

1.Install related dependent libraries:

```
sudo apt-get install -y util-linux procps hostapd iproute2 iw haveged dnsmasq
```

2.Before turning on the hotspot, your Raspberry Pi cannot be connected to WIFI, and the WIFI module cannot be turned off, so when you test the hotspot function, you need to connect the necessary peripherals for the Raspberry Pi.

3.Under normal circumstances, if the robot program is not connected to the WIFI when it is turned on, it will automatically turn on the hotspot. You can use your phone or computer to search for the WIF named Adeept_Robot. The default password is 12345678. Once the connection is successful, you can log in to 192.168 .12.1: 5000 using a browser to open the WEB application to control the robot.

4.If your Raspberry Pi is connected to peripherals, and you want to test the Raspberry Pi 's ability to turn on hotspots, you can click the WIFI icon in the upper right corner of the Raspberry Pi 's desktop, click the name of the connected WIFI, click forget, and never turn Off WIFI, if it is already in the off state, you need to turn it on.

5.When the WIFI module of the Raspberry Pi is turned on and is not connected to any known network, you can enter the following command on the console to turn on the WIFI:

```
sudo killall python3
```

```
sudo create_ap wlan0 eth0 Adeept_Robot 12345678
```

"Adeept_Robot" is the name of the WIFI hotspot, "12345678" is the password of the WIFI hotspot.

## 2.5 Q&A

- Run "webServer.py" error**: Could not start camera.**

The Raspberry Pi will automatically run the webServer.py program. The webServer.py program will occupy the camera. If you run the camera-related program again, the program will report an error. When you need to run the program manually, you need to stop the automatic running program.

1. Enter in the Raspberry Pi command line:

**sudo killall python3** (stop the automatic running program)

2. Enter the Raspberry Pi camera command: "**raspistill -t 1000 -o image.jpg**". Normally, the Raspberry Pi will take a photo named image.jpg, which can be viewed by entering "**ls**". If an error is reported, please check whether the camera is installed correctly until this command can run successfully. Or the camera is damaged.

Note: The Raspberry Pi needs to be powered off when plugging or unplugging the camera cable, otherwise the camera may be damaged.

3. Enter "**sudo python3 adeept_picar-b/server/webServer.py**" in the Raspberry Pi

4. If there is no error reported when running the program, please use Google Chrome to enter the web control interface through "Raspberry Pi IP address" + ":5000". At this time, you can see the Raspberry Pi camera screen on the web control interface. For example: "192.168.3.98:5000".

- Run "webServer.py" error: **remote I/O error.**

It may be because i2c is not turned on.

Enter in the Raspberry Pi:

**sudo raspi-config**

Select "3 Interface options"

Select "P5 I2C"

Select "Yes" to open i2c

- Run "webServer.py" error: "*ImprotError: xxxxx failed to import*".

It may be caused by the unsuccessful installation of the dependent library. Please try to reinstall the dependent libraries. Enter in the Raspberry Pi command line:

**sudo python3 adeept_picar-b/setup.py**

If the above method does not work, please try to burn the image file we provided again. The dependent library has been installed in the image file and can be used after configuring WiFi and SSH.

The image file is located in the "Adeept picar-b Image File" PDF linked below.

Link: https://www.adeept.com/learn/detail-38.html

- Run "webServer.py" error: "*RuntimeError: ws2811_init failed with code -5.*"

Please use "sudo" to run the program.

- Run "webServer.py" error:

*"RuntimeError：ws2811_init failed with code -3(Hardware revision is not supported)"*

Enter in the Raspberry Pi:

**sudo pip3 install rpi_ws281x**

**sudo pip3 install rpi-ws281x --upgrade**

- Errors occur with `permission denied` prompt when I manually run `server.py` or `webServer.py`.

The Raspberry Pi needs the root permission to run the dependent libraries for WS2812 LED lights control.

You need to add `sudo` to the beginning of `server.py` or `webServer.py` to run the program. [PATH] is the product folder path.

E.g: **sudo python3 adeept_picar-b/server/webserver.py**

sudo python3 [PATH]/server.py

sudo python3 [PATH]/webServer.py

- Raspberry Pi can't boot.

Remove all parts on the driver board. Only connect the board to Raspberry Pi and power supply, reboot.

Observe if the green light of the Raspberry Pi is blinking.

If the indicator does not flash or the indicator flashes at the same interval, it means that the image file has not been burned successfully, please re-burn the system.

- "Remote side unexpectedly closed network connection" shows on a popup window.

There can be error prompts during installation because the Raspberry Pi will auto reboot after the installation, which will disconnect the board.

- Raspberry Pi failed to connect a WiFi.

Please connect to the Raspberry Pi through the mouse, keyboard, monitor, and then click the wifi icon in the upper right corner of the Raspberry Pi interface to enter the wifi information.

- I can not connect to the Raspberry Pi terminal via SSH

Please connect to the Raspberry Pi through the mouse, keyboard, monitor, and then click on the main menu in the upper left corner of the Raspberry Pi interface, select "preferences", select "Raspberry Pi Configuration", select "Interfaces", set the SSH bar to "Enabled", then Click "OK"

- SSH can't connect, error: WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!

Enter the following in the command line and press Enter

ssh-keygen -R <the Raspberry Pi's IP address>

For example:

```
ssh-keygen -R 192.168.3.157
```

Then you can SSH to the Raspberry Pi again

- Can I power the Robot HAT and Raspberry Pi via USB?

A 2A output is required for a Raspberry Pi 3B, when at least 3A is needed for a Raspberry Pi 4. You can use the USB power for software installation and testing, but it's not suitable for high power module like servo or motor adjustment as it may result in low voltage. It's recommended to use battery for power here.

- no cv2 error occurs when I manually run `server.py` or `webServer.py`.

OpenCV is not installed correctly. Type in the command sudo pip3 install opencv-contrib-python in the Raspberry Pi to manually install OpenCV.

- When using a computer to copy ssh and wpa_supplicant.conf to the SD card, it prompts that there is no SD card

  If this happens, unplug the card reader and connect it to the computer.